# Reading Between the Lines

How People Perceive and Detect Cultural Erasure in AI-Retold Bengali Folktales

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# When Culture Meets Computation

Bengali folktales encode cosmological beliefs, moral frameworks, and regional identities that cannot be reduced to surface plot. When LLMs retell these stories, they do not produce neutral versions — they produce retellings shaped by Western defaults encoded in their training data. This study asks: **can people tell the difference?**



# Three Interlocking Research Questions

1

## RQ1 — Erasure Patterns

What types of cultural erasure do LLMs produce in Bengali folktale retellings, and how do they vary across model, prompt strategy, and language?

2

## RQ2 — Human Detection

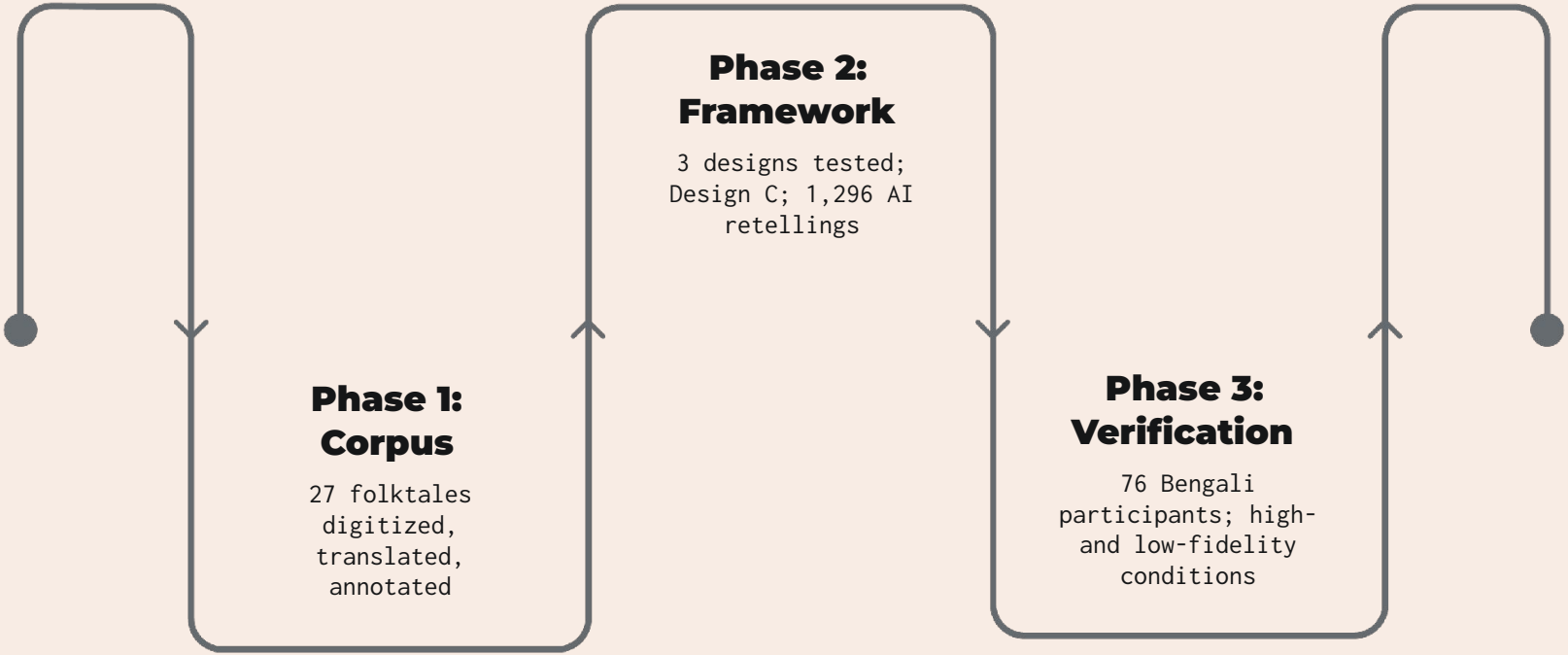
Can Bengali-speaking participants detect AI-generated retellings? What variables — distortion severity, folklore familiarity, age — mediate detection?

3

## RQ3 — Framework Alignment

How do automated Cultural Fidelity Perception Framework scores correlate with human perceptual detection of cultural erasure?

# Three-Phase Research Methodology




Each phase produces validated outputs that serve as inputs to the next, ensuring rigorous, behaviorally-grounded evaluation at every stage.

# Design Philosophy

Our research design centers on responsible, rigorous evaluation of machine-generated folklore retellings. The following four principles guided every stage — from corpus construction through evaluation and interpretation.

### 1. Diagnostic Intent

Design evaluation metrics to reveal specific failure modes (lexical loss, narrative shifts, socio-cultural substitutions) rather than producing single aggregated scores.

### 2. Human Primacy

Prioritize human judgment for culturally embedded readings: native speakers and folklore experts validated ground truth and assessed perceptual salience of erasures.

### 3. Transparency

Document models, prompts, translation steps, and annotation protocols openly so results are reproducible and interpretable by digital humanities peers.

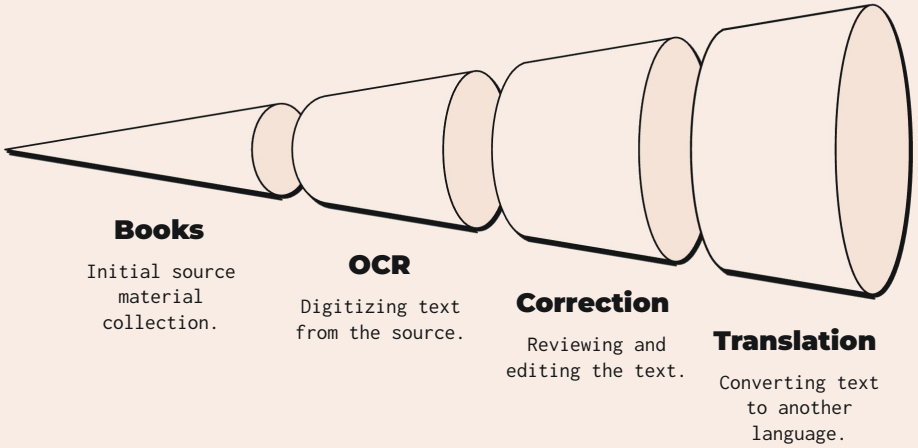
### 4. Cultural Humility

Acknowledge limits of automated methods, center community knowledge, and avoid imposing external narrative norms when judging fidelity.

# Corpus Construction — Pipeline

We assembled a rigorously curated corpus to support cross-linguistic and model-comparative evaluation.






Result: 54 validated stories (after human editing and benchmarking) used as base material for generation and evaluation.

# Translation Quality Assurance

To create faithful English retellings we compared three translation models and applied a mixed automated + human protocol.



## Models Compared

- Helsinki's Opus MT
- BUET's BanglaT5
- Gemini 2.5 Pro



## Evaluation & Selection

We used MQM evaluation to assess adequacy and fluency. The top-performing model was selected for bulk retelling and followed by human post-editing focused on cultural terms and narrative voice.



## Human Post-Editing

Editors corrected cultural referents, idioms, and socio-pragmatic cues to preserve local meaning while maintaining readability in English.

# Experimental Scale

The experiment was designed to explore model × prompt × language effects with sufficient variety to detect interaction patterns.

1

### Stories

27 stories for the main experimental set (10 from Thakurmar Jhuli; 17 regional tales).

2

### Languages

2 languages (Bangla, English)

3

### Models

- Llama-3.3-70B-Instruct
- GPT-OSS-120B
- Qwen-2.5-7B-Instruct
- DeepSeek-R1-0528
- Gemini-2.0-Flash-Exp
- Mistral Devstral-2512

4

### Prompts

Six distinct prompt strategies (see next card) designed to probe retention and transformation of cultural elements.

# Prompt Design

We crafted prompts to manipulate the context the model receives and to test for both subtle and overt forms of cultural alteration.

## Prompt 1 — Simple-Control

Neutral instruction: "Retell the following story in the same language with natural fluency."

## Prompt 2 — Context-Rich

Provides cultural context, character notes, and setting specifics to encourage fidelity.

## Prompt 3 — Moral-Constraint

Asks model to preserve moral lessons and culturally specific ending motifs.

## Prompt 4 — Fail-Safe

Instructs the model to flag uncertainties and avoid inventing cultural facts when unsure.

# Proppian Inversion and SDI

We adapt Proppian function analysis as a lens to detect structural inversions (e.g., role swaps, addition of non existent-functions) introduced by models. The SDI (Structural Domination Index) flags instances where Western narrative templates supplant indigenous structures.



## Proppian Inversion

Identify addition or modification of existing functions (e.g., magical helpers, altered quest logic, happily ever after).

## SDI

Quantifies the degree to which narrative voice, moral framing, or plot causality shifts toward Western norms.

# The Dimensions of Cultural Fidelity

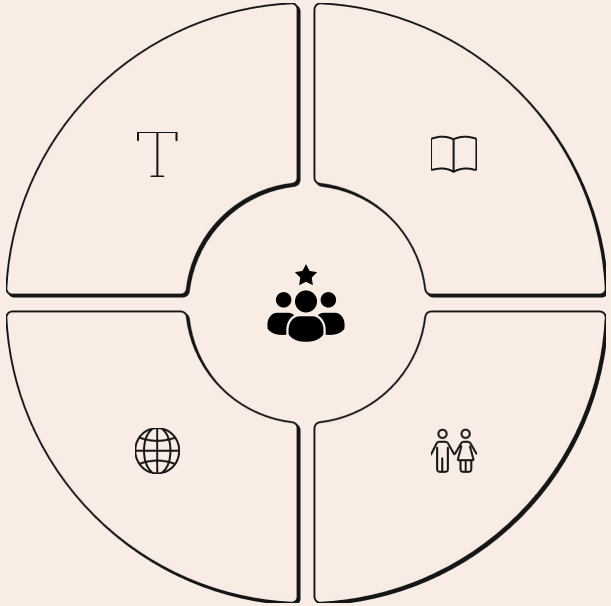
Our Cultural Fidelity Perception Framework decomposes fidelity into measurable dimensions that capture lexical, narrative, and socio-cultural aspects.

## Lexical Integrity

Did culturally irreplaceable words survive? (proper names, ritual terms, kinship vocabulary)

## SDI — Structural Domination Index

Was a Western narrative imposed or did local narrative structures predominate?



## Narrative Fidelity

Did plot functions, sequence, and character roles remain intact?

## Socio-Cultural Integrity

Did behaviors, norms, and value signals survive translation and retelling?

# Evaluating Cultural Fidelity: Three Designs

| Design A — Lexical Scorer | Design B — Structural Alignment | Design C — Holistic Auditor ✓ |
| --- | --- | --- |
| Fuzzy matching on culturally-specific terms. Deterministic and interpretable — but captures only one dimension of cultural fidelity. | Cosine similarity via multilingual embeddings. **Refuted:** scored 0.85 similarity on a story where all cultural content was replaced with Western equivalents. | Gemini 2.5 Flash with 4-dimension rubric and hard constraint overrides. **Selected:** Pearson r = 0.90–0.96 against human expert judgments. |

# Human Study Methodology

### Part 1: High-Fidelity Retellings

Participants read stories that closely preserved the original narrative, characters, and cultural details. After each story, they judged whether it was **human-written** or **AI-generated**.

### Part 2: Low-Fidelity Retellings

Participants then read stories in which cultural content had been weakened, generalized, or replaced. They made the same human-vs-AI judgment to test whether fidelity changes affected detection.

### 1. Read the story

Each participant reviewed a story excerpt presented in Bengali.

### 2. Classify the source

Participants indicated whether the story was human-written or AI-generated.

### 3. Rate confidence

After each judgment, participants reported confidence on a **1–5 scale**.

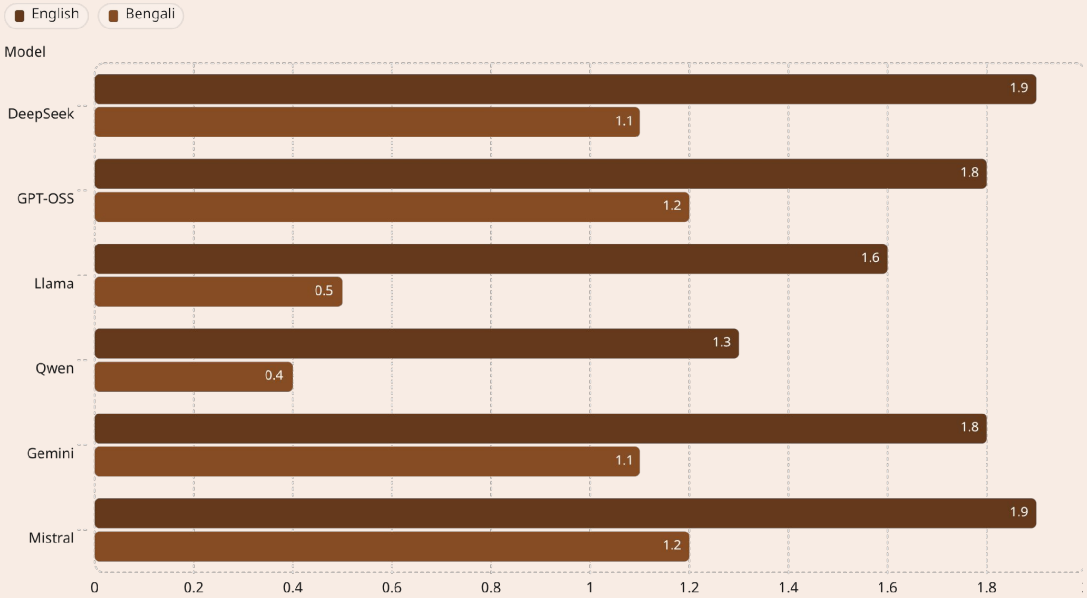
## Collected Prior to Main Task

- Consent
- Non-identifiable demographic factors
- Full disclosure of inclusion of AI generated content

## Done After the Main Task

- Assurance to withdraw
- Gratitude
- All entries protected and kept safe

DATA ANALYSIS · CULTURAL FIDELITY




# Cultural Scores by Model: English vs. Bengali

Summary: Every evaluated model shows markedly lower cultural-fidelity when prompted in Bengali. English prompts produce the highest mean scores (still ≤2/5); Bengali prompts reduce scores dramatically. Qwen collapses to near-zero (SGF dominance), indicating near-complete loss of cultural markers in Bengali retellings.

### Key takeaway

Language of interaction is a primary determinant of cultural preservation across models.

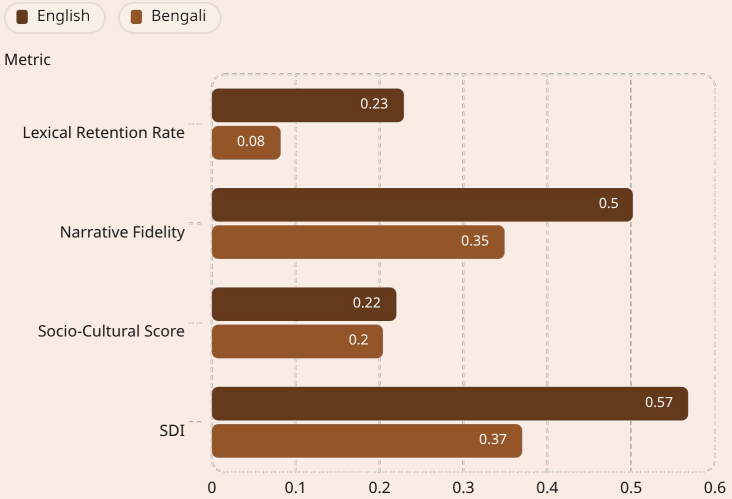
### Implication

Model training or failure modes disproportionately erase minority-language cultural content.

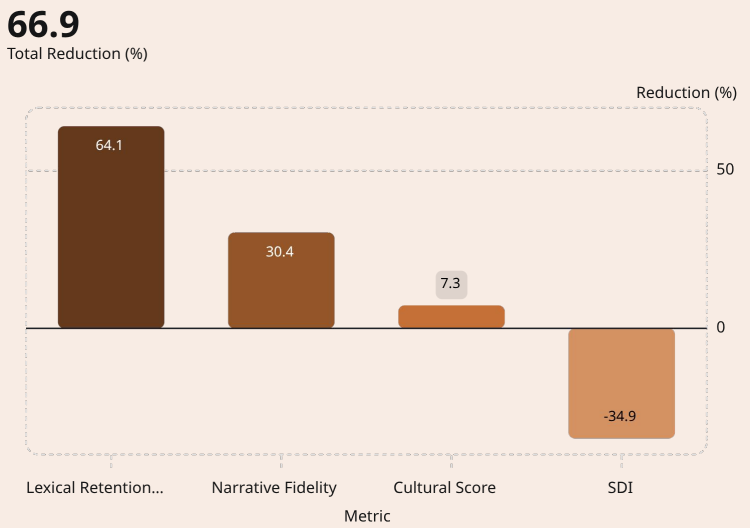
DATA ANALYSIS · LANGUAGE EFFECT

# Bengali Prompting Makes Erasure Worse

Quantified impact: switching English→Bengali reduces lexical retention by 64.1%, narrative fidelity by 30.4%, and increases SDI by ~35%. Language-of-interaction is not a neutral variable — it systematically amplifies cultural loss in model outputs.




**Absolute Values** — English vs. Bengali

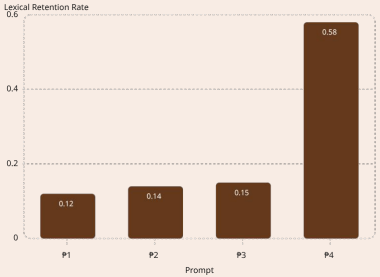



**Percentage Reduction** — English→Bengali

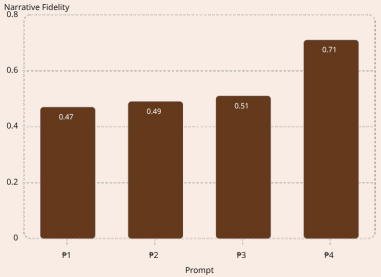
DATA ANALYSIS · PROMPT EFFECT

# The Prompt Makes the Difference — But Only P4 Helps

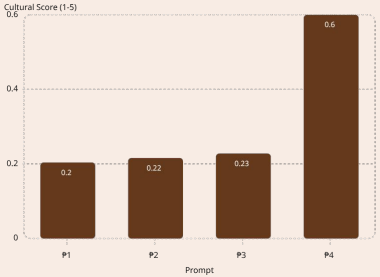
Across lexical retention, narrative fidelity, cultural score, and SDI, prompts P1–P3 yield statistically indistinguishable low cultural outcomes. Only P4 (full source + explicit fail-safe instruction) meaningfully improves cultural score — yet SDI (structural displacement) remains elevated, indicating persistent Westernizing tendencies even when surface cues improve.



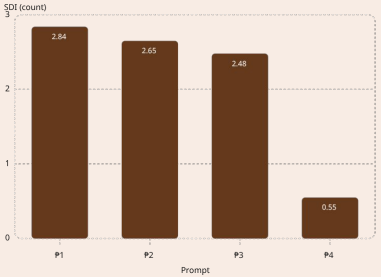

**Lexical Retention Rate**




**Narrative Fidelity**




**Socio-Cultural Score**




**SDI (count)**

### P4 vs others

- P4 preserves more lexical items and narrative motifs.
- SDI still signals structural inversion despite lexical gains.
- Practical: prompt engineering alone is insufficient for full fidelity.

### P1–P3

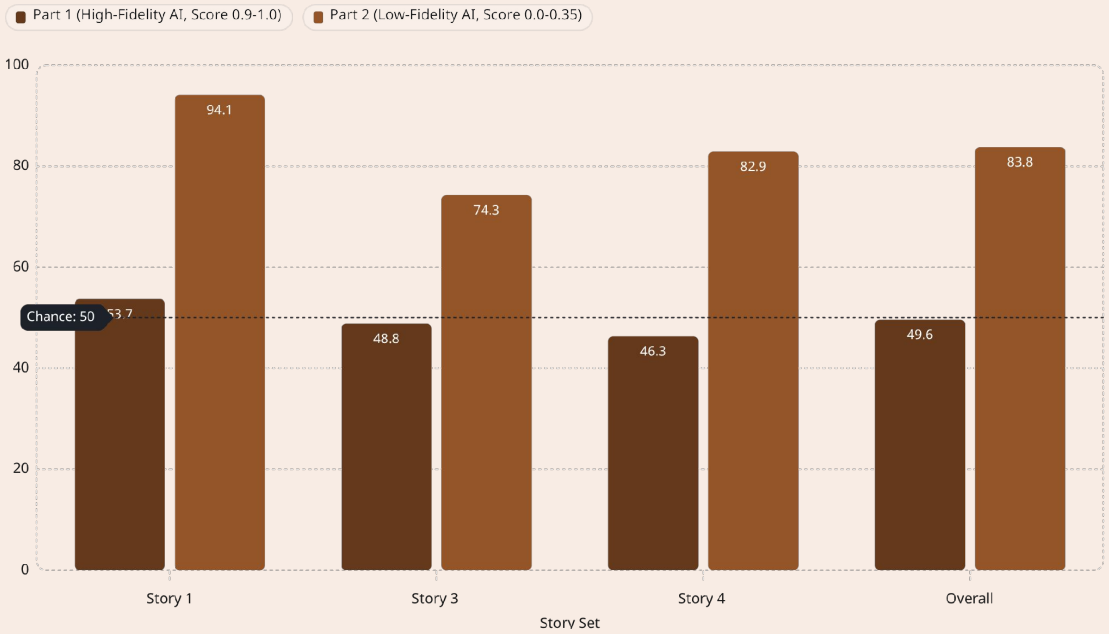
Low cultural retention; fails to surface culturally specific forms.

### P4

Best among tested prompts, but not a full remedy — structural issues persist.

HUMAN STUDY · DETECTION RESULTS

# The Paradox of Fluency



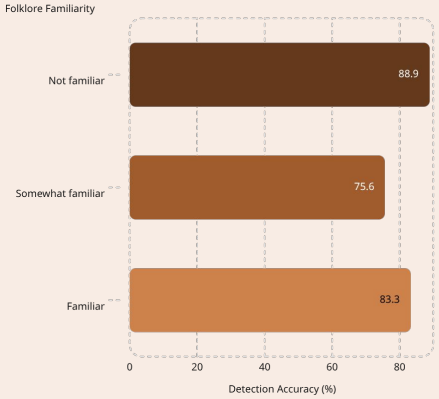

Part 1 (high-fidelity outputs): human detection ≈49.6% (chance). Part 2 (low-fidelity AI): detection ≈83.8%. When AI preserves cultural form, humans fail to flag AI authorship; fluency and cultural fidelity combine to create an effective concealment.

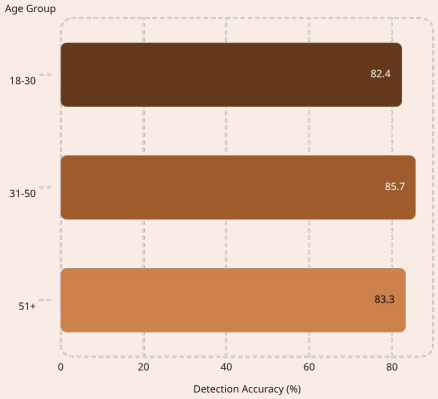
Critical gap: human review alone cannot catch culturally faithful AI-generated content. Policy and tooling must evolve to detect what readers cannot.

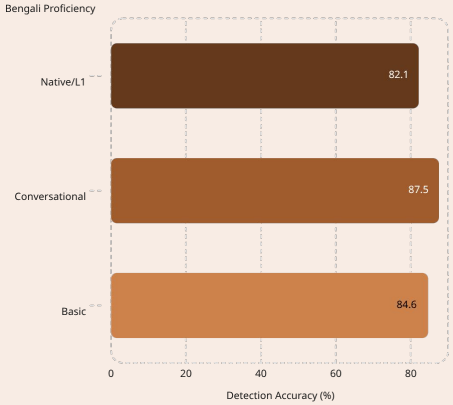
HUMAN STUDY · DEMOGRAPHICS

# Who Detects Cultural Erasure Best?

Across folklore familiarity, age groups, and Bengali proficiency, Part 2 detection exceeds ~75% in every subgroup. Differences are modest — cultural erasure becomes perceptible once present, regardless of deep domain expertise.



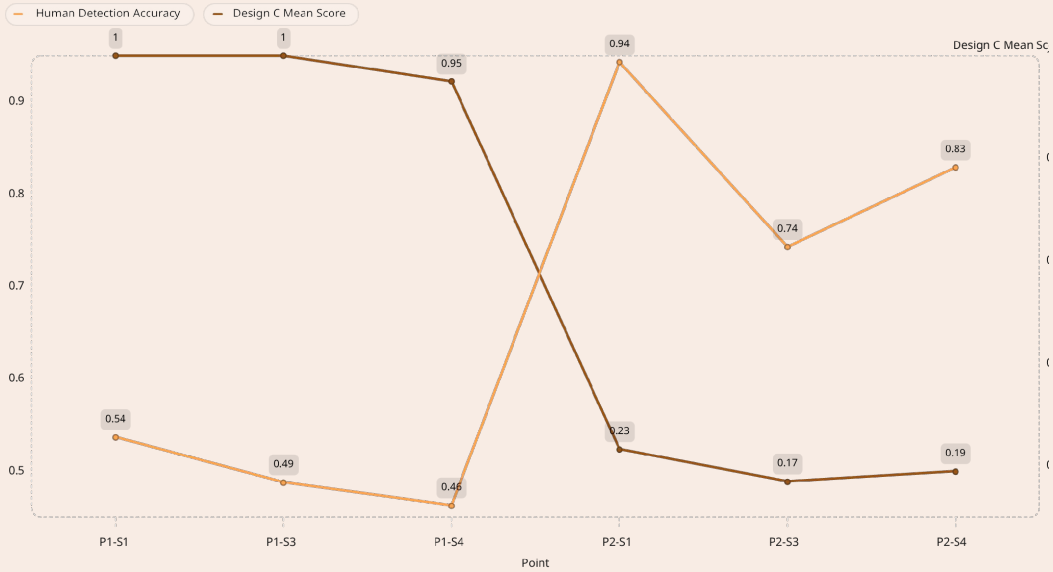

## Uniformity

Detection performance shows limited variance by demographic subgroup.

## Interpretation

Once visible, erasure cues are broadly detectable; subtle erasure that preserves fluency remains invisible.

FRAMEWORK VALIDITY · RQ3




Source:

# Our Framework Predicts Human Detection Perfectly

Spearman ρ = −1.00 indicates perfect rank-order alignment between the Design C cultural score (CFPF) and human detection accuracy across evaluated conditions. This validates CFPF as a reliable, automatable proxy for human expert judgment.

## Why this matters

- Enables scalable cultural audits without requiring human panels for every test.
- Supports reproducible, JSON-anchored scoring for research and governance.

## Limitations

- Alignment measured here is rank-order; absolute scores still need expert interpretation.
- Further cross-cultural validation required before generalizing beyond Bengali folktales.

QUALITATIVE & CALIBRATION ANALYSIS

# What Participants Noticed

Thematic coding reveals consistent qualitative cues tied to perceived authenticity: language register, absent cultural markers, narrative logic shifts, formulaic phrase omission, and archetype drift. Brier score analysis shows confident miscalibration—participants were often certain about authenticity despite being wrong.

- **Language register**

  "Too formal / too conversational for a folktale."

- **Missing markers**

  "No রাক্ষস, no পুণ্যি — felt generic."

- **Narrative logic**

  "Endings felt tidy; formulaic openings missing."

- **Calibration failure**

  High confidence on incorrect Part 1 judgements → risk for downstream trust.

## The Double Bind

# When fidelity hides authorship

Two simultaneous facts create a critical vulnerability: (1) humans cannot reliably detect AI when cultural fidelity is high (detection ≈50%), and (2) models systematically erase Bengali cultural content, especially when prompted in Bengali. The consequence: cultural erasure can be widespread yet invisible to readers and reviewers.

| **Reader** | **System** |
| --- | --- |
| The reader cannot reliably protect cultural integrity through judgement alone. | The current generation of models cannot reliably preserve culturally specific structures and markers. |

⚠ Cultural erasure is most dangerous when it is indistinguishable from authentic writing.

CONTRIBUTIONS

# What This Study Adds

## First systematic audit

Comprehensive evaluation across 6 models, 4 prompts, and 2 interaction languages — fills a major gap in non-English cultural AI research.

## CFPF (novel)

A validated, JSON-anchored Cultural Fidelity Perception Framework with four cultural dimensions, designed for reuse and auditability.

## SDI (new metric)

Structural Displacement Indicator operationalizes Proppian inversion, quantifying narrative Westernization in retellings.

## Human-model alignment

Empirical evidence (Spearman $\rho = -1.00$) that CFPF rank-orders outputs identically to human detection — enabling scalable auditing.

FUTURE WORK

# Where This Research Should Go Next

- Cross-cultural validation: apply CFPF to Yoruba, Tamil, Navajo, Andean oral traditions.
- Real-time CFPF integration: build a pipeline for live evaluation at generation time.
- Fine-tuning interventions: use annotated corpus to reduce SDI and restore native narrative structures.
- Larger human studies: increase sample size for robust calibration/Brier analyses and subgroup power.
- Policy & disclosure: develop cultural-impact disclosure standards for AI in literary/educational contexts.
- Tooling: provide accessible auditor dashboards for practitioners and community stakeholders.

Next steps emphasize both technical mitigation (fine-tuning, pipelines) and governance (disclosure, community-driven audits).

# Summary of Findings

Concise synthesis of the study's core results on cultural fidelity in LLMs.

### RQ1 — Structural Erasure

Bengali prompting intensifies cultural erasure across all LLMs. Rich prompts (P4) help but cannot remove structural displacement.

### RQ2 — Human Detection

Bengali speakers detect culturally unfaithful AI with 83.8% accuracy (Part 2), but accuracy drops to chance (49.6%) when AI outputs are culturally fluent (Part 1).

### RQ3 — CFPF Design C

Design C achieves perfect rank-order alignment with human detection ($\rho = -1.00$), validating it as a scalable proxy for expert cultural evaluation.

Conclusion: Cultural loss is structural, measurable, and often invisible to unwarned readers.



# *The stories survive.*

## *The question is whether what survives is still the story.*

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## Thank You

Questions & Discussion Welcome